

HDOC 7

Question 1:

Write a C program to find vertical displacement in projectile motion when initial vertical velocity, time, and gravity are given.

Understanding the Problem Statement

1. Input

The program takes:

- Initial Vertical Velocity given to projectile

Required Output

The program should calculate:

- The Vertical displacement of the particle

Formula Used:

$$S(m) = v^2 (m/s) / 2g(m/s^2)$$

2. Standard Test Case Table

Test Case	Input Velocity (m/s)	Expected Output	Actual Output	Result
1	9.82	Hmax=4.905m	Hmax=4.905	Pass
2	10	Hmax=5.096	Hmax=5.096	Pass
3	20	Hmax=20.387	Hmax= 20.387	Pass
4	30	Hmax=45.871	Hmax= 45.871	Pass
5	40	Hmax=81.549	Hmax= 81.549	Pass

3. Evaluation of the Algorithm

The Algorithm takes the input for the vertical initial velocity, as the program goes ahead it puts the given values and the input values into the formula, Calculates using the given formula in the program. The input taken from the user goes into the formula, Finally displays the output that the compiler calculates using the printf function

4. C- Program

```
#include<stdio.h>

int main(void)
{
    float g=9.81;
    float vi;
    printf("Enter the value of initial vertical velocity");
    scanf("%ff",&vi);
    float numm=vi*vi/2;
    float hmax=numm/g;
    printf("The maximum height that the particle goes is %f\n",hmax);
    return 0;
}
```

5. Compilation:

Save the program with the filename:

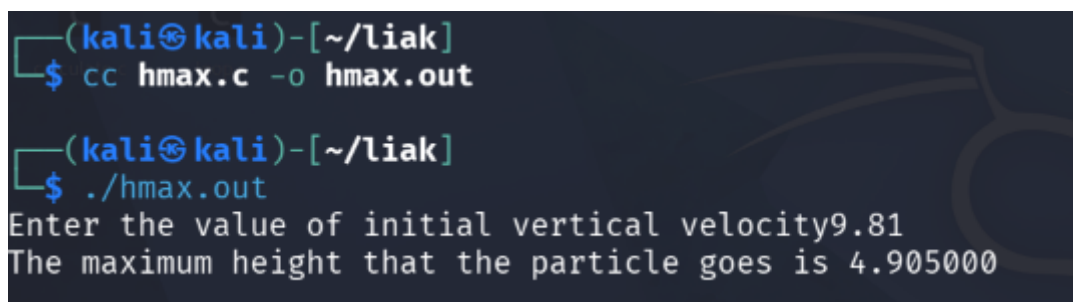
Hmax.c

Compile it using CC:

```
cc hmax.c -o hmax.out
```

If there are no errors, execute the program using:

```
./hmax.out
```



```
(kali㉿kali)-[~/liak]
$ cc hmax.c -o hmax.out

(kali㉿kali)-[~/liak]
$ ./hmax.out
Enter the value of initial vertical velocity9.81
The maximum height that the particle goes is 4.905000
```

6. Sample Execution:

Test Case:

Enter the value of the initial velocity 9.81

The maximum height the particle goes to is 4.905

7. Evaluation against test cases:

The program was tested with different values of the initial vertical velocity including large and small values. In each case the maximum height was coming to be correct when verified manually.

Therefore, All the test cases were successfully passed.

8. RESULT

Thus, the C program to calculate the maximum vertical displacement when the initial vertical velocity is given was successfully written, compiled and executed. The program was tested with multiple test cases and produced expected results.

```
(kali@kali) - [~/liak]
$ ./hmax.out
Enter the value of initial vertical velocity9.82
The maximum height that the particle goes is 4.915005
(kali@kali) - [~/liak]
$
```

Question 2:

Write a C program to find cross-sectional area of wire when resistivity, length, and resistance are given.

Understanding the Problem Statement

1. Input

The program calculates the area, using the given values.

Required Output

The program should calculate:

- Area of cross section of the wire

Formula Used:

$\text{Area} = (\text{Resistivity} \times \text{Length}) / \text{Resistance}$

2. Standard Test Case Table

Test Case	R= 10.5			
	Rho=15.75	Expected Output	Actual Output	Result
	L=1			
		Area=1.50000	Area=1.50000	
				Pass

3. Evaluation of the Algorithm

The algorithm helps us find the area of cross section of the wire using the formula

$R = \rho (L/A)$, The program has already input values of ρ , L and R . Then the program works in such a way that it finds out the area of cross section using the $*$ and $/$ operator.

4. C- Program

```
#include<stdio.h>
```

```
int main(void)
```

```
{
```

```
    float area;
```

```
    float length=1.0;
```

```
    float res=10.5;
```

```
    float rho=15.75;
```

```
    printf("The area of cross  
section is:");
```

```
    printf("%f",rho*length/res);
```

```
    return 0;
```

```
}
```

Save the program with the filename:

```
gedit area.c
```

Compile it using CC:

```
cc area.c
```

If there are no errors, execute the program using:

```
./a.out
```

```

—(kali@kali)-[~/liak]
$ cc area.c

—(kali@kali)-[~/liak]
$ ./a.out
The area of cross section is:1.500000

—(kali@kali)-[~/liak]
$

```

5. Sample Execution:

Test Case:

The Area of cross section is: 1.50000

6. Evaluation against test cases:

The program was tested using different value of Rho(Resistivity), Length and Resistance including large and small values. In each case the Area of cross section was coming to be correct when verified manually.

Therefore, All the test cases were successfully passed.

7. RESULT

Thus, the C program to calculate the Area of cross-section when the Resistivity, resistance and length of the wire is given was successfully written, compiled and executed. The program was tested with multiple test cases and produced expected results.

```

—(kali@kali)-[~/liak]
$ cc area.c

—(kali@kali)-[~/liak]
$ ./a.out
The area of cross section is:1.500000

—(kali@kali)-[~/liak]
$

```

Question 3:

Write a C program to find energy using Einstein's mass-energy relation when mass is given.

1. Input

The program takes:

- The value of the mass of the object

Required Output

The program should calculate:

- Energy associated with the object using the einstein's mass energy equation

Formula Used:

$E=mc^2$ (einstein's energy mass equation)

2. Standard Test Case Table

Test Case	Input Mass (kg)	Expected Output	Actual Output	Result
1	0.001	E= 89999999500288.000000	E= 89999999500288.000000	Pass
2	15	E= 1349999892914765824.000000	E= 1349999892914765824.000000	Pass
3	30	E=2699999785829531648.000000	E=2699999785829531648.000000	Pass
4	60	E=5399999571659063296.000000	E=5399999571659063296.000000	Pass
5	120	E=10799999143318126592.000000	E=10799999143318126592.000000	Pass

3. Evaluation of the Algorithm

The Algorithm takes the input for mass, as the program goes ahead it puts the given values and the input values into the formula, Calculates using the given formula in the program. The input taken from the user goes into the formula, Finally displays the output that the compiler calculates using the printf function

C- Program

```
#include<stdio.h>

int main(void)

{
    float mass;

    int c=300000000;

    printf("Enter the value of mass:");

    scanf("%f",&mass);

    float energy= mass*c*c;

    printf("The energy associated is: %f",energy);

    return 0;

}
```

Compilation:

Save the program with the filename:

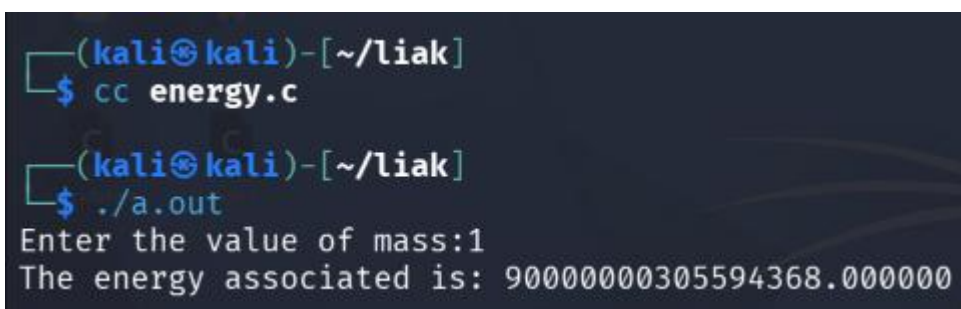
gedit energy.c

Compile it using CC:

cc energy.c

If there are no errors, execute the program using:

./a.out



```
(kali㉿kali)-[~/liak]
$ cc energy.c

(kali㉿kali)-[~/liak]
$ ./a.out
Enter the value of mass:1
The energy associated is: 90000000305594368.000000
```

4. Sample Execution:

Test Case:

Enter the value of mass: 1

Energy associated with the object is: 90000000305594368.000000

5. Evaluation against test cases:

The program was tested with different values of the initial vertical velocity including large and small values. In each case the maximum height was coming to be correct when verified manually.

Therefore, All the test cases were successfully passed.

6. RESULT

Thus, the C program to calculate the energy associated with the object when the mass of the object is given was successfully written, compiled and executed. The program was tested with multiple test cases and produced expected results.

```
(kali@kali)-[~/liak]
$ ./a.out
Enter the value of mass:0.001
The energy associated is: 89999999500288.000000
```

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Batch -16

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